Islamic University of Technology (IUT)

CSE 4614: Technical Report Writing

Topics Covered in the LaTeX Learning Section

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April 07, 2024

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1 Lab 1: Introduction to the Basics

The following topics are covered (not chronologically) in the LaTeX learning section:

- Structure of a LaTeX source code
  - The `documentclass` command
  - The preamble section
  - The `document` environment
  - Structure of a LaTeX command and environment
    - General style of positional (mandatory) and optional arguments
- Inclusion of packages
  - Geometry package to change the paper type, margin, and layout of document
- Options for the `article` class to change font size and paper type
- Changing fonts by importing respective packages
- Use of `lipsum` package and command to generate filler text for type-setting
- Setting title, author, and date information and generating title with the `maketitle` command
  - Creating a separate title page with the `titlepage` environment
- Creating paragraphs and lines with line and paragraph breaks with the \, `newline` and `par` commands respectively
- Moving to new page with the help of `newpage` and `pagebreak` commands (there are slight differences, students can explore themselves)
- Making text bold, italic, and underlined
- Emphasizing text with the `emph` command and its behavior in the `article` class
- Single and multi-line comments (with the help of `comment` environment from the `verbatim` package)
- Changing text sizes, e.g. large, Large, Huge, small, tiny, etc.
- Creating numbered and unnumbered (asterisk (*) version of commands) sections, subsections, and sub-subsections
- The `abstract` environment
- Centering text with the `center` environment (students should explore making text left or right aligned)
- Creating unordered (`itemize`) and ordered (`enumerate`) lists and nesting them
- Generating table of contents
- Hyperlinking items (basic) with the help of `hyperref` package

2 Lab 2: Mathematical Expressions

The following topics are covered (not chronologically) in the LaTeX learning section:

- The two mathematical modes/environments (how to and their differences):
  1. Inline mode, with:
     - `\(...\)`
     - `$...$`
     - `\begin{math}...\end{math}`
  2. Display mode, with:
     - `\[...\]`
     - `$$...$$`
     - `\begin{displaymath}...\end{displaymath}`
     - `\begin{equation}...\end{equation}` *(numbered)*
- The `amsmath` package and the `equation*` environment (all later `amsmath` environments have `*` variants)
- Difference between mathematical functions, operators, operands, and text in mathematical environments
- Mathematical commands, operators, symbols, such as `times`, super and subscripts, `frac`, `sin`, `cos`, `int`, `sum`, `prod`, `lim`, `limits` (to position lower and upper limits vertically)
- Equality, non-equality, greater than, less than, greater equal, less equal symbols
- `left` and `right` commands to match bracket pairs and `.` for open brackets
- Greek letters, `mathcal`, `mathfrac`, `mathbb` from `amssymb` (also show multi-import in a single `usepackage` command)
- Infinity (`infty`), arrows, and spacing with `quad`, `qquad`, and likes of `\.`
- `mathbf` and `text` commands
- Referencing an equation, first with vanilla `ref` and then with the help of `autoref` command from `hyperref` package
- Vector symbol and center dot (`cdot`) for dot product symbol
- Different `amsmath` environments and aligning equations:
  - `multiline`, `split` (inside another math environment), `align`, `gather`
- Different types of matrices with the help of `matrix`, `bmatrix`, `Bmatrix`, `pmatrix`, `vmatrix`, `Vmatrix` etc. environments

3 Lab 3: Floats and Figures

The following topics are covered (not chronologically) in the LaTeX learning section:

- The `graphicx` package to include graphics
- Using `\graphicspath{ {./folder1/} {./folder2/} }` command to set path to graphics directory (multiple directories are separated with inner curly braces)
- `\includegraphics{path}` command to insert a basic image
  - The convention of not writing file types to automatically decide best type if multiple are available (like using `\DeclareGraphicsExtensions{.pdf,.png}`) for optimizing workflow in production environment
  - Why `pdfs` are best choices for good quality report
- Optional arguments in `\includegraphics` to:
  - change the height and width (both absolute units using `px`, `cm`, etc. and relative units like `\linewidth`, `\textheight`, etc.) while maintaining (changing either height or width) or breaking (changing both height and width) aspect ratio.
  - change the scale (uniform) of the image
  - rotate the image
- Introduction to floats and their behavior
- The `figure` environment:
  - `h`, `t`, `b`, and `p` placement modifiers
  - `H` modifier using `float` package
  - Setting up and positioning the caption
  - Using label to refer to the figure and the necessity of a caption for labelling (if `\ref` and `\autoref` commands from `hyperref` package was not taught in the last lab, it would be taught here)
- Wrapping text around figures using `wrapfigure` environment from the `wrapfig` package
- Inserting subfigures using `subcaption` package and `subfigure` environment, referring to individual figures inside a subfigure and the subfigure as a whole
- Generating list of figures (like table of contents)
- Existence of packages for sideway figures and captions
- Existence of `floatrow` package for greater control of floats

4 Lab 4: Tables

The following topics are covered (not chronologically) in the LaTeX learning section:

- Basic tables without any package using the `tabular` environment with vertical and horizontal separators and extra space after row `\\[<HEIGHT>>]`
- Utilizing basic column types, such as `l`, `r`, and `c` for auto-sized columns
- Creating fixed sized column with the help of `p{<WIDTH>}` column type and using `\newline` inside it
- Positing tables as float using the `table` environment with table caption and label for reference
- Changing global properties of tables:
  - Rule width using `\setlength{\arrayrulewidth}{<WIDTH>}`
  - Column separator using `\setlength{\tabcolsep}{<WIDTH>}`
  - Row height stretch using `\renewcommand{\arraystretch}{<SCALAR MULTIPLIER>}`
- Introduction to the `array` package
  - `m` and `b` column type for vertically middle and bottom aligning cell contents respectively
  - Customizing column types within cell and in main column type via modifier injection and the `\arraybackslash` command
  - Creating custom column type
- Creating fixed-sized and auto distributing tables using `tabularx` package
- Creating multi-column and multi-row (using `multirow` package) cells and the `\cline` command
- Combining multi-column and multi-row (multi-row is inner)
- Introduction to the `booktabs` package and booktabs style tables
- Multi-page long table using the `longtable` package
- List of tables (in contents)
- Existence of other packages related to tables and changing table color

5 Lab 5: Bibliography and Citations

The following topics are covered (not chronologically) in the LaTeX learning section:

- How the bibliography backend works
- Preparing the database (`.bib`): books, articles, proceedings, software, wiki
- Citation using just `bibtex`:
  - Single-citation
  - Multi-citation
  - Citation style and printing
- Extending capability with the help of the package `biblatex`:
  - Benefits and reason for using automated citations
  - Setting up the preamble with style, sorting, and resources with the help of the command: `\addbibresource[<options>]{<resource>}`
  - Necessity of resource extension
  - Adding multiple resource and existence of glob matching
  - Citing in different styles:
    * Normal citations
    * Name of authors with the help of `\textcite{<key>>}`
    * Parenthesized citation with the help of `\parencite{<key>>}`
    * Footnote citation with the help of `\footcite{<key>>}`
    * Multi-citation
  - Printing bibliography, changing title, and placing in ToC with the help of the command: `\printbibliography[heading={bibintoc}, title={Bibliography}]`.
  - Citation in draft style